

# Distributed Deep Learning on GPU-based Clusters

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# Welcome!

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**Are you interested in training/fine-tuning or using deep learning models for inference on a networked cluster of GPUs?**

- If the answer is yes, you have come to the right tutorial.

# Tutorial attendees will learn ...

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- Different forms of parallelism for deep learning
- Setting up a training (fine-tuning) job on GPGPUs
- Different parallel deep learning frameworks and how to use them
- Using a model in inference mode on multiple GPGPUs

# Deep learning

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- An area of machine learning that uses artificial neural networks to learn complex functions
  - Often from high-dimensional data: text, images, audio, ...
- Widespread use in computer vision, natural language processing, etc.

# Tutorial organizers

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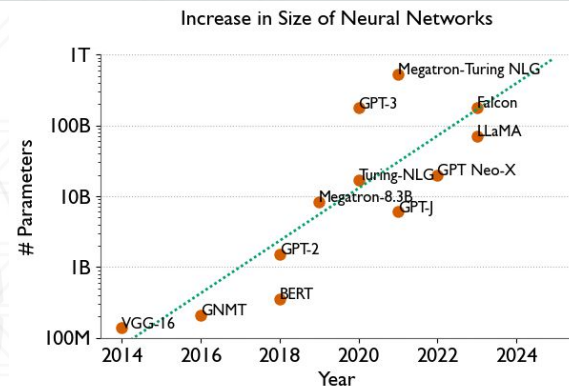
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# Parallel / distributed training

- Many opportunities for exploiting parallelism
- Iterative process of training (epochs)
- Many iterations per epoch (mini-batches)
- Many layers in DNNs



| Framework      | Type of Parallelism | Largest Accelerator Count | Largest Trained Network (No. of Parameters) |
|----------------|---------------------|---------------------------|---------------------------------------------|
| FlexFlow       | Hybrid              | 64 GPUs                   | 24M*                                        |
| PipeDream**    | Inter-Layer         | 16 GPUs                   | 138M                                        |
| DDP**          | Data                | 256 GPUs                  | 345M                                        |
| GPipe          | Inter-Layer         | 8 GPUs                    | 557M                                        |
| MeshTensorFlow | Intra-Layer         | 512-core TPUv2            | 4.9B                                        |
| Megatron**     | Intra-Layer         | 512 GPUs                  | 8.3B                                        |
| TorchGPipe**   | Inter-Layer         | 8 GPUs                    | 15.8B                                       |
| KARMA          | Data                | 2048 GPUs                 | 17B                                         |
| LBANN**        | Data                | 3072 CPUs                 | 78.6B                                       |
| ZeRO**         | Data                | 400 GPUs                  | 100B                                        |
| ZeRO-Infinity  | Data                | 512 GPUs                  | 32T                                         |
| AxoNN          | Inter-Layer         | 384 GPUs                  | 100B                                        |

# Materials and Accounts

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- Tutorial link:

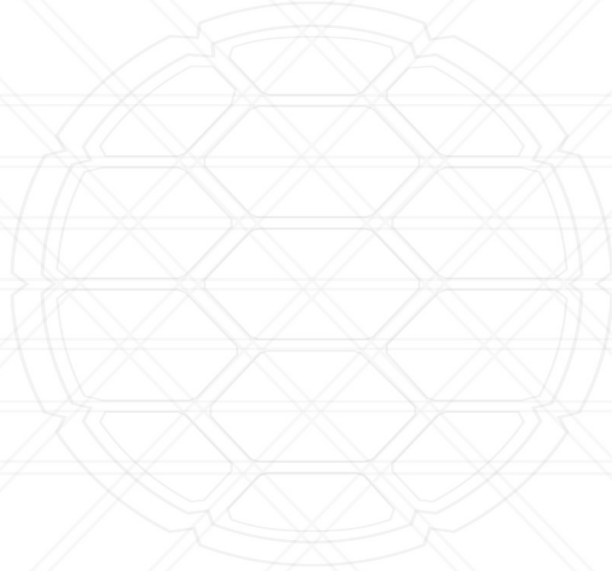
<https://github.com/axonn-ai/distrib-dl-tutorial>



- Accounts: <https://forms.gle/KNgImwkWb6yWHARv8>
- ssh <username>@login.zaratan.umd.edu







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